

National Aeronautics and
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EXPLORE PROCUREMENT

The cornerstone of NASA's current and future missions

Ames Facilities Support Services II Industry Day



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RELIABILITY CENTERED MAINTENANCE & TIERED MAINTENANCE - OVERVIEW

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NASA's facilities maintenance and operation philosophy is to support NASA's mission by proactively pursuing techniques to provide safe, sustainable, efficient, and reliable facilities.

- **NPR 8831.2F: Facilities Maintenance and Operations Management**
 - NID 8831.124: Interim directive for Condition Based Maintenance (CBM)

Reliability Centered Maintenance (RCM) strategy is to determine the most applicable cost-effective maintenance strategy to minimize risk and failure.

Tiered Maintenance (TM) classifies facilities and assets into hierarchical tiers based on their impact on operations.



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RELIABILITY CENTERED MAINTENANCE

RCM is the process that is used to determine the most effective approach to maintenance. It involves identifying actions that, when taken, will reduce the probability of failure and that are the most cost effective. It seeks the optimal mix of Condition-Based Actions, other Time- or Cycle-Based actions, or a Run-to-Failure approach. (See also Condition-Based Maintenance, Predictive Testing & Inspection.)

RCM is **a continuous process** that requires sustainment throughout the life cycle of a system, utilizes data from the results achieved, and **feeds this data back to improve design and future maintenance**.

RCM's aim is to **optimize maintenance strategies**

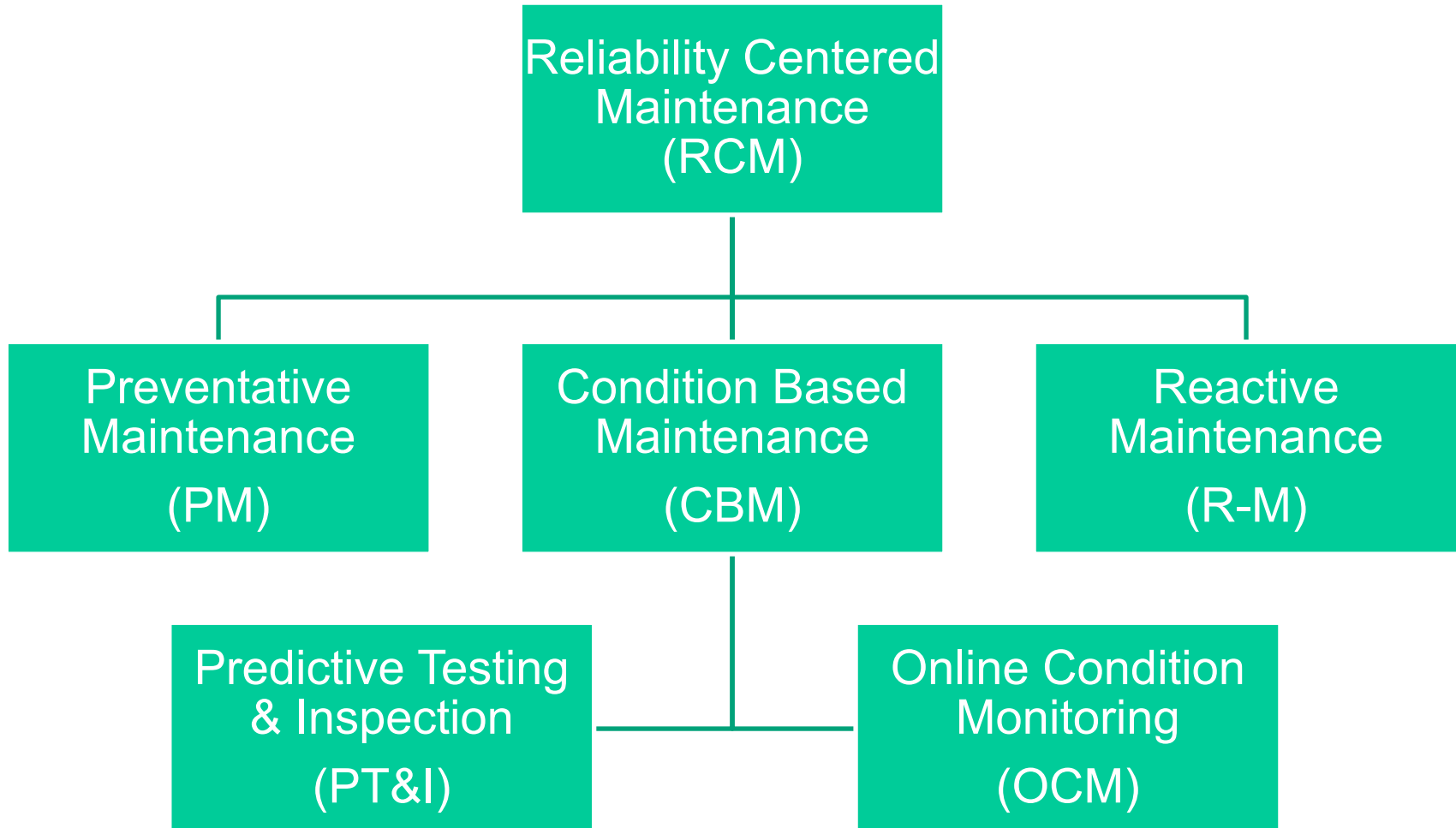
- Prioritizing critical assets
- A more detailed approach by analyzing each asset's function and failure modes
- Determines the most effective maintenance strategy
- Reduce reactive Unscheduled Maintenance, Increase Proactive/Planned Maintenance



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Maintenance Hierarchy





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TIERED MAINTENANCE

Tiered maintenance (TM) is a strategic, **engineering-based approach** that classifies facilities into **hierarchical tiers**.

- Mission-driven
- Tiered maintenance offers a structured approach by categorizing **facilities based on their mission relevancy**, allowing for more strategic planning and resource allocation.
- **Three key aspects of tiered maintenance:**
 1. Facility Tier Rating (Mission Relevancy: MR)
 2. Equipment Asset Criticality Rating (EACA)
 3. Maintenance Planning

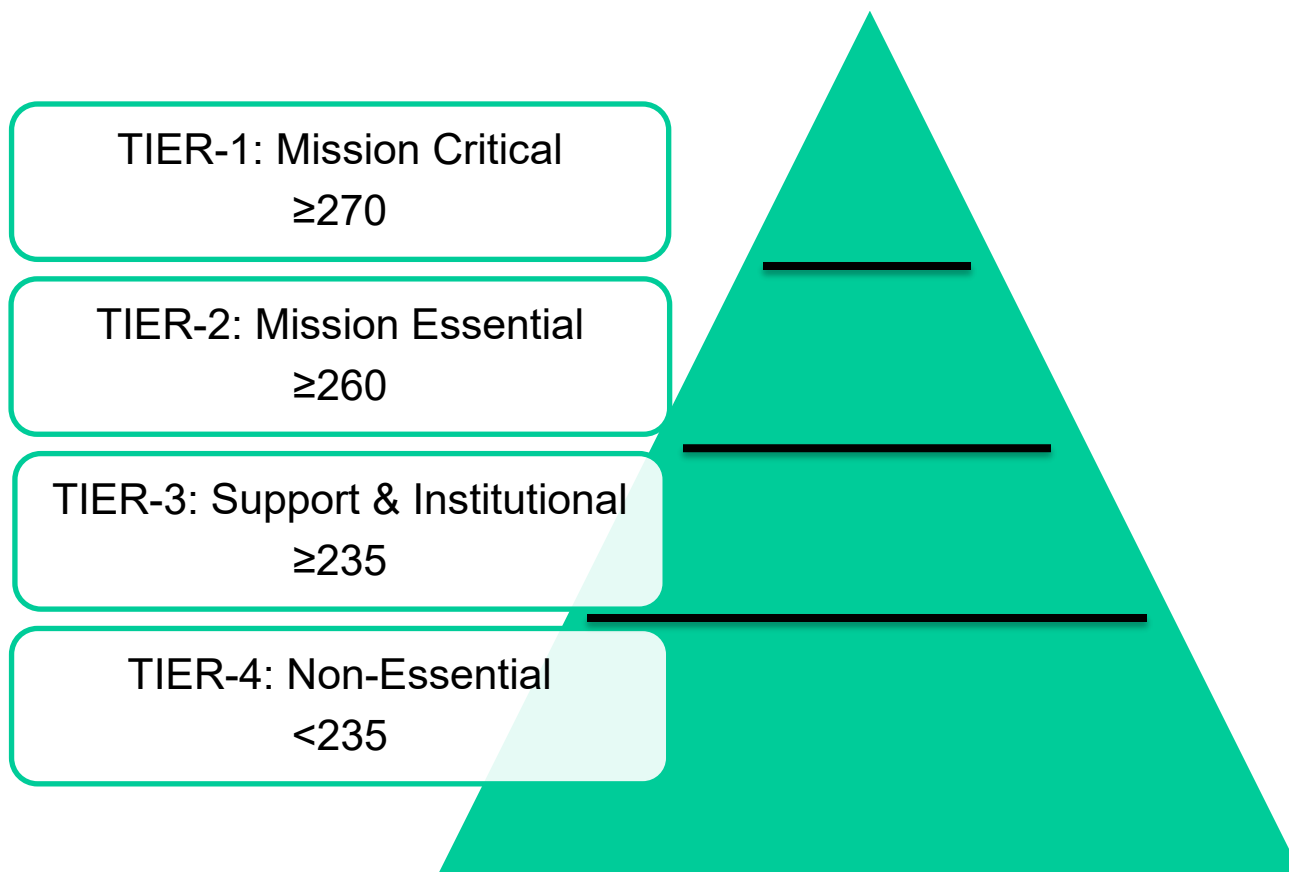


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1. Facility Tiering

Mission Relevancy Score (Max 280)





2. Equipment Asset Criticality Assessment (EACA)

The EACA is a quantitative process to determine how assets affect mission success based on five qualitative categories.

Collateral equipment criticality level ranked from 1 (highest) to 4 (lowest).

Safety

- Level of harm to personnel

Environmental

- Impact on environment/regulatory compliance

Mission

- Mission relevancy based on Master Planning

Schedule

- Time to restore functionality

Cost

- Cost to repair or replace (restore capability)



3. Maintenance Planning

Equipment Criticality Level

Level 1

- CBM
- Frequent PMs
- Critical spare parts
- Root Cause Analysis (RCA)

Level 2

- Limited CBM
- ❖ Selective PT&I
- Focused PMs
- Some spare parts

Level 3

- Limited CBM
- Primarily routine PMs
- Outsourced spare parts

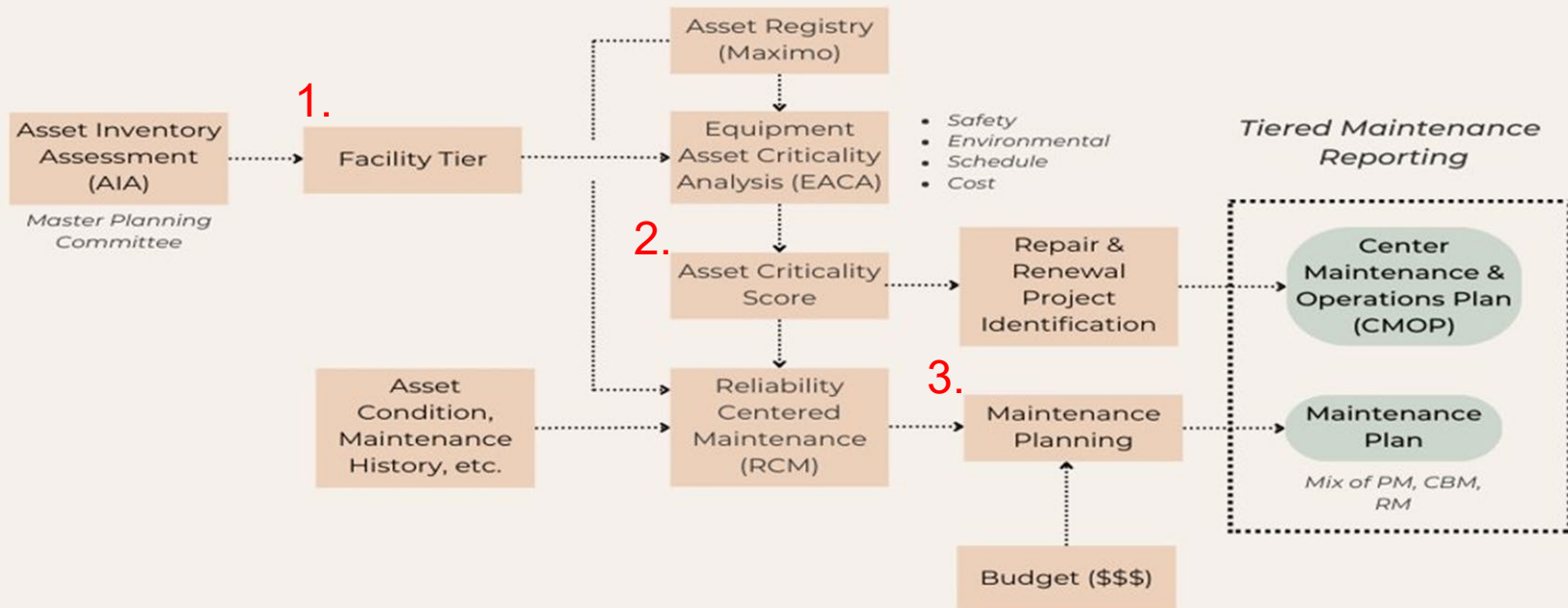
Level 4

- Limited PMs
- Increased R-M
- Repairing facility assets after discovery of failure.



TIERED MAINTENANCE

is to optimize and focus scarce resources to maintain the most mission relevant facilities and collateral equipment needed to support the center's mission.





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COMPLIANCE VERIFICATION

Compliance and maintenance optimization is measured through various monthly and annual metrics reports:

- Monthly Tiered Maintenance Report
- Center Maintenance and Operations Plan (CMOP)
 - Annual and 5-Year Work Plan
 - Annual CBM investments
 - PT&I Finds/Cost Avoidance
 - Annual End of Year Maintenance Metrics